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September 24, 2026

Re: Revised submission DATABASE-2026-0054 - “KinaseCanvas: An Integrated Resource for Curated and Novel Kinase-Substrate Interactions”

Dear Database Editors,

Thank you for the opportunity to revise our manuscript describing **KinaseCanvas**, an open-access resource that expands the functional annotation of the understudied “dark” kinome by integrating kinome-wide specificity atlases with curated kinase-substrate interactions. We thank both reviewers for their careful reading and have addressed every comment; a point-by-point response is enclosed, and changes to the manuscript are shown in blue.

Principal revisions:

- **Corrected pipeline.** The reviewers’ checks led us to re-audit the pipeline and correct two implementation errors: an identifier-resolution error and an error in the functional-score cut-off. The resource has been regenerated with added consistency checks, and every figure and statistic reflects the corrected pipeline.
- **New analyses.** Threshold sensitivity, a continuous confidence score for every interaction, the information content and independence of the evidence layers, and coverage by tissue specificity.
- **Explorer.** Substrate-first queries, a pair mode that ranks candidate sites with their flanking sequences, quantitative evidence for every interaction, and a detail panel linking to UniProt and PubMed.
- **Data access.** Every released interaction carries the evidence values behind its classification; the tables, code and full prediction matrix are archived on Zenodo.

Key results of the revised resource:

- **Network expansion: 21,916 novel kinase-substrate pairs**, for 30,280 pairs in total across 426 kinases and 3,901 substrates.
- **Illuminating the dark kinome:** The share of interactions involving understudied kinases rises 4.6-fold, from 3.4% to 15.7%.
- **Biological utility:** Layering localization and functional evidence onto motif scores prioritises CDK1 among five motif-compatible kinases for the mitotic phosphosite RANGAP1 S442, consistent with the study that first reported the site.

Enclosed: the response to reviewers; the revised manuscript with changes in blue, and a clean copy; and Supplementary Tables S2-S4 and the relaxed motif tier (Table S1, 3.6 GB, is on Zenodo).

Thank you for your time and consideration.

Sincerely,

Chinmaya Joisa

Postdoctoral Fellow, on behalf of all authors